

The empirical recurrence for OEIS A283200: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A283200 records “Empirical recurrence of order 96”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 121$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 96, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 96 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A283200 is “Number of nX5 0..1 arrays with no 1 equal to more than two of its horizontal and vertical neighbors, with the exception of exactly one element.”. It has offset 1 and begins

0, 176, 7899, 263232, 7842272, 218628378, 5855079969, 152494818512, ...

The entry states:

Empirical recurrence of order 96 (see link above)

As of the “Last modified” line on the live entry (Mar 02 2017 20:42:45, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1744$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 121$ of the 1744, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 121$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 242$ exact terms of the model returns order 96 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{96} c_i a(n-i),$$

c_1	28	c_{25}	−234769043566892	c_{49}	7403428894208152	c_{73}	134853628066
c_2	10	c_{26}	1580616598874967	c_{50}	3721529330640498	c_{74}	−3335679279
c_3	168	c_{27}	808682924583426	c_{51}	−1804609736505548	c_{75}	38649526894
c_4	−41041	c_{28}	−3445053830831286	c_{52}	−4217557555016143	c_{76}	−14299396714
c_5	−430820	c_{29}	−2244238492980354	c_{53}	−2085914316462446	c_{77}	−239231186
c_6	−4362982	c_{30}	6392270955178144	c_{54}	198811192802172	c_{78}	52737676
c_7	−24903108	c_{31}	5229011248610108	c_{55}	1472796750036122	c_{79}	−339805288
c_8	−122483638	c_{32}	−10052167770291945	c_{56}	611000691869098	c_{80}	−280367457
c_9	−309153232	c_{33}	−10380695914918890	c_{57}	245334479343046	c_{81}	−263209508
c_{10}	−275759792	c_{34}	13142656259043066	c_{58}	−67866731382884	c_{82}	108291219
c_{11}	3899826336	c_{35}	17603114522637820	c_{59}	23200447592	c_{83}	−14286404
c_{12}	17319492983	c_{36}	−13569720378498060	c_{60}	30528784081614	c_{84}	5956638
c_{13}	40070301018	c_{37}	−25326997770311584	c_{61}	−30184529814332	c_{85}	−2126910
c_{14}	−58536885576	c_{38}	9356971881580514	c_{62}	−12472975650614	c_{86}	2248948
c_{15}	−454818289702	c_{39}	30264383783227960	c_{63}	−61746398216042	c_{87}	716760
c_{16}	−815494146134	c_{40}	−504203947322433	c_{64}	−19574196903107	c_{88}	−374353
c_{17}	1769882896652	c_{41}	−28616146199776064	c_{65}	−16976584906746	c_{89}	36000
c_{18}	8944677837174	c_{42}	−9622243017302441	c_{66}	−465157080588	c_{90}	−12112
c_{19}	−3337964840426	c_{43}	19070115957036632	c_{67}	−2645677518574	c_{91}	5176
c_{20}	−49374541672346	c_{44}	15375353209113490	c_{68}	−1994467026565	c_{92}	−3608
c_{21}	−2230241731506	c_{45}	−5151296833013398	c_{69}	330716538010	c_{93}	−1056
c_{22}	195663114963928	c_{46}	−12762062913148842	c_{70}	70677646390	c_{94}	560
c_{23}	47783785895558	c_{47}	−5630623275971374	c_{71}	596964964074	c_{95}	32
c_{24}	−612278320180868	c_{48}	3745796269251654	c_{72}	−173507143714	c_{96}	−16

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 96, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $96 + 4$ terms removed returns order 96 as well, so $\deg q_{\min} = 96 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $96 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 96 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A283200.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.